

Sujay Karthikeyan

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Skills

Languages Java, C++ , Python, Linux, Git, AWS
CAD Inventor, Creo, Onshape, SolidWorks

Software Microsoft Office, ROS2, OpenCV, ROS,
Gazebo, Matlab, Arduino, SLAM, Simulink

Experience

SmartNet Lab

July 2025 – Present

Research Assistant

- Developed a high-fidelity simulation environment for autonomous drone swarms using **ROS2 (Jazzy)** and **Gazebo Harmonic on Ubuntu 24.04**.
- Engineered a scalable multi-agent launch system capable of deploying three independent networked UAVs simultaneously with decentralized control logic.
- Achieved real-time operational telemetry with odometry updates of **100Hz**, validating system stability for cooperative path-planning missions.

Autodesk

Aug 2024 – Present

Design and Make Ambassador

- Host interactive workshops to help students and professionals refine their AutoCAD skills
- Offer guidance and resources to support aspiring designers and engineers
- Cultivate a community of innovators using Autodesk tools to solve real-world challenge
- Connected with Autodesk Employees on improving engineering software at the University of North Carolina at Charlotte

InspireNC

Jul 2023 – Present

Director

- Made a yearly budget for Non-Profit, analyzing trends from previous years
- Conducted networking events with local companies such as Stemarga and Accenture to learn business strategy
- Raised 35K over 6 months through grants, fundraisers, and donations
- Hosted Workshops regarding fundraising and business strategy to newly established non-profits
- Partnered with Metlife, US Air Force, Lenovo to spread engineering education to children

Education

University of North Carolina at Charlotte

Aug 2024 – May 2027

Bachelor of Science in Mechanical Engineering and Physics

Relevant Coursework

- Math and Science for Engineers, Calculus III, Thermodynamics, Statics, Differential Equations
- CAD, Matlab, Simulink, Environmental Science, Computer Science (Java)

Certifications

- Autodesk Inventor - Certified User
- Autodesk AutoCAD - Certified User
- Introduction to Aerospace Structures and Materials - Delft University of Technology

Projects

Autonomous Tilt-Rotor VTOL UAV Design

Dec 2025

- Engineered a custom **2-axis tilt-rotor mechanism** capable of 90-degree transition for VTOL operations; designed interference-free sub-assemblies integrating bearings, servos, and carbon fiber structural elements
- Optimized the airframe for **Design for Additive Manufacturing (DFAM)**, utilizing shelling, offset split-planes, and tolerance analysis to ensure successful FDM printing of thin-walled structural components.

Forge – AI-Powered 3D Design Platform

September 2025

- **Developed a browser-based CAD application** that enables users to generate editable 3D models from natural language prompts, utilizing the **Google Gemini API** for intent recognition and geometric parameter extraction
- **Implemented an STL export pipeline**, creating a custom STL Exporter module that converts browser-generated geometry into manufacturing-ready files for 3D printing.
- **Designed a responsive UI with HTML5 and Tailwind CSS**, focusing on a chat-centric workflow ("Design at the Speed of Thought") that lowers the barrier to entry for CAD tools